TERM PAPER:

Here's a simple breakdown of the rubric of our term paper:

Details

Please submit your term paper as a single PDF file. Each group only needs one submission.

Grading rubric:

Item	Points	What we're looking for
Motivating the problem	10	A good explanation to why the problem is worth solving/studying (e.g., what bad/good things might happen).
Design & Methodology	15	A reasonable approach to solve/study the problem using credible tools and/or data sources.
Critical Analysis	15	Reasonable interpretation and reflection on the results, plus original insights and/or lessons learned.
Technical Accuracy	15	A correct and accurate explanation of the technical concepts.
Depth of Research	15	A thorough investigation of the topic within the defined scope.
Citation & References	10	Proper citation of relevant work that helps to illustrate the ideas or build the arguments.
Structure & Organization	10	Logical progression/connections between sections (the so-called flow).
Readability	5	Overall readability of the paper.
Available & Quality of Artifacts	5	Artifacts available with proper documentation.
Total	*100*	

What We're Actually Doing (Plain English)

The Big Question: When an IoT device (like a security camera or router) is setting up a secure connection, will it *choose* a weak/outdated security method if given the option — even when better options exist?

Step 1: Collect our targets Find Internet-exposed IoT devices (cameras, routers, NAS devices) using tools like Shodan or Censys. Also grab a comparison set of regular web servers so we have a baseline.

Step 2: Build 3 client "personalities" We pretend to be different types of clients connecting to these devices:

- 🟢 **Modern client** — only offers strong, up-to-date cipher suites
- 🟡 **Mixed client** — offers both strong AND weak cipher suites (like a slightly outdated device would)
- 🔴 **Legacy client** — only offers old, weak cipher suites

Step 3: Connect to each device with all 3 personalities Record exactly what the server picks each time. Did it choose the strongest option available? Did it pick a weak one even when a strong one was on the table?

Step 4: Measure the results Track:

- Which TLS version got negotiated
- Which cipher suite was selected
- Whether forward secrecy was used
- Whether the server enforces its *own* preference order or just accepts whatever the client wants

Step 5: Compare IoT vs. regular web servers Are IoT devices worse at this than normal websites? By how much?

The Key Insight We're Testing:

Most studies just check *"does this device support weak ciphers?"* — we're going further and asking *"will it actually USE them?"* Those are very different things. A device might technically support TLS 1.3 but still happily negotiate TLS 1.0 with an old client. That's the exposure we're measuring.

What We're NOT doing (important for the paper): We are NOT simulating a hacker intercepting connections. There's no man-in-the-middle. We're just acting like a real-world misconfigured or outdated client and seeing how servers respond. Think of it as: *"how badly does this server behave when talking to a dumb client?"*